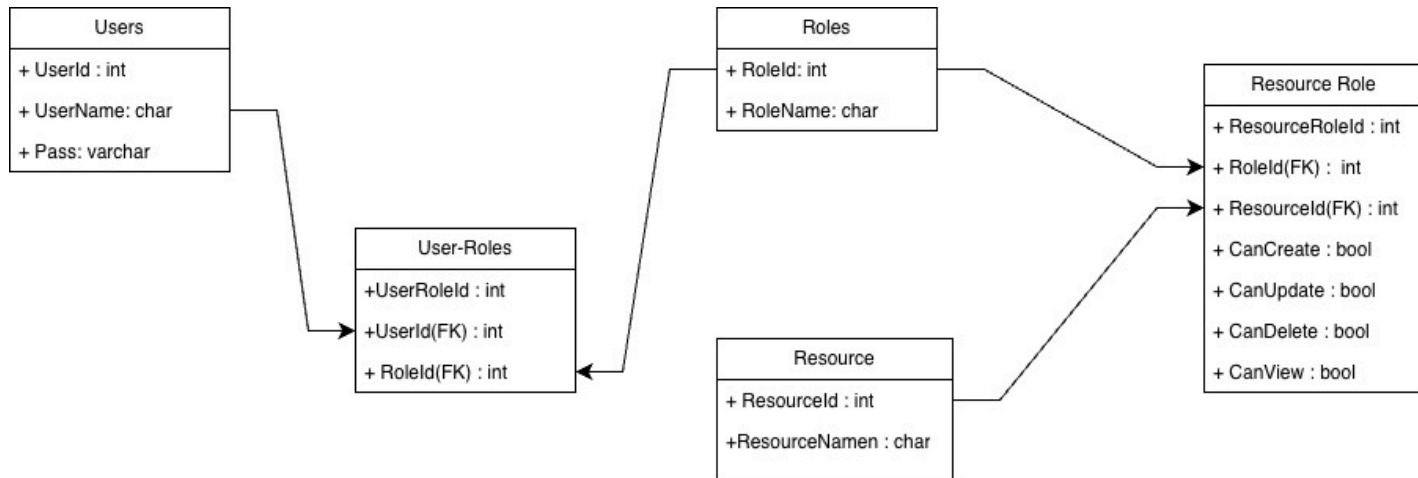


Data Base Design

Role Based Access Control Class Diagram:



Example:

Users Table

ID	UserName	Pass(Hashed)
101	Hitlar	Joajnfeqr ger
201	Putin	nfnaenvaenvaev
301	Kim	nvaernveqnv

Roles Table

RoleId	RoleName
1	Principle
2	Teacher
3	Parents

UserRoles Table:

UserRoleId	UserId	RoleId
11	101	1
22	101	2
33	201	2
44	301	3

Resources Table:

ResourceId	ResourceName
1	Result
2	Account
3	Attendance
4	Library

ResourceRole Table:

ID	RoleId	Resource Id	Can View	Can Create	Can Update	Can Delete
1	1Principle	1(Result)	Yes	Yes	Yes	Yes
2	1Principle	2(Account)	Yes	Yes	Yes	Yes
3	2Teacher	1(Result)	Yes	Yes	Yes	NO
4	2Teacher	2(Account)	Yes	NO	NO	NO

Example Query:

For Hitler ID is 101

```
SELECT
    res.ResourceId, -- Use ID for logic
    res.ResourceName, -- Use Name for display
    MAX(rr.CanView) AS View,
    MAX(rr.CanCreate) AS Create_New,
    MAX(rr.CanUpdate) AS Update_Existing,
    MAX(rr.CanDelete) AS Delete_Data
FROM Users u
JOIN UserRoles ur ON u.ID = ur.UserId
JOIN ResourceRole rr ON ur.RoleId = rr.RoleId
JOIN Resources res ON rr.ResourceId = res.ResourceId
WHERE u.ID = 101 -- Search for Hitler by his ID
GROUP BY res.ResourceId, res.ResourceName;
```